

ZANJAN, Iran, Islamic Republic of

WMO: 407290

Lat: 36.6598N

Lon: 48.5218E

Elev: 1663

StdP: 82.88

Time Zone: 3.50 (IRN)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.7	-10.8	-17.7	1.0	-9.0	-15.3	1.2	-6.0	12.1	0.8	10.5	3.1	1.5	90	0.526

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.2	35.1	16.1	33.7	15.9	32.1	15.8	18.5	31.0	17.7	29.8	16.9	28.8	4.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
14.0	12.2	23.7	13.2	11.6	23.0	12.4	11.0	22.5	59.7	30.8	56.6	29.8	54.0	28.8	23.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.1	10.1	8.3	-17.6	37.2	4.0	1.6	-20.5	38.4	-22.9	39.3	-25.2	40.2	-28.1	41.4		
(5)			-17.8	19.8	4.0	1.7	-20.7	21.0	-23.0	22.0	-25.2	22.9	-28.1	24.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.9	-1.1	1.0	5.9	10.9	16.0	21.1	24.1	23.9	19.2	13.3	6.1	1.5		
	DBStd	9.43	4.80	4.43	4.21	3.61	2.89	2.87	2.71	2.22	2.90	3.26	3.78	3.96		
	HDD10.0	1164	344	254	138	31	1	0	0	0	0	9	124	264		
	HDD18.3	2859	602	487	384	224	82	8	1	1	23	158	368	522		
	CDD10.0	1851	0	1	12	57	187	332	437	431	276	111	7	1		
	CDD18.3	505	0	0	0	1	11	90	180	173	49	2	0	0		
	CDH23.3	6496	0	0	1	14	189	1182	2117	2142	787	64	0	0		
	CDH26.7	2852	0	0	0	1	27	470	1039	1047	263	4	0	0		
(14) Wind	WSAvg	3.2	2.9	3.1	3.5	3.3	3.2	3.4	3.7	3.5	3.1	2.7	2.7	2.7		
(15) Precipitation	PrecAvg	294	28	30	38	53	37	9	8	4	3	19	37	28		
	PrecMax	403	63	54	93	106	80	50	34	22	19	58	127	66		
	PrecMin	170	10	8	2	7	4	0	0	0	0	0	0	1		
	PrecStd	53	13	12	21	25	21	12	9	6	5	16	31	17		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.7	14.2	21.9	25.3	29.1	35.4	37.8	36.5	33.0	27.1	18.9	15.0		
		MCWB	4.8	6.6	9.2	11.8	13.2	15.3	16.8	15.9	15.4	12.8	8.4	6.3		
	2%	DB	9.1	11.8	18.6	22.2	27.2	32.9	35.8	34.9	31.2	25.1	16.8	11.5		
		MCWB	3.4	5.0	7.6	10.4	12.9	15.5	16.6	15.8	14.8	11.6	7.6	4.9		
	5%	DB	7.2	9.5	15.9	20.2	25.6	31.1	33.9	33.5	29.8	23.2	14.8	9.3		
		MCWB	2.4	3.8	6.7	9.4	12.3	14.7	16.4	15.7	14.2	10.9	6.9	3.9		
	10%	DB	5.6	7.6	13.7	18.3	23.8	29.4	32.1	32.1	28.0	21.2	12.6	7.4		
		MCWB	1.7	2.9	5.7	8.8	11.8	14.4	16.3	15.6	13.6	10.3	6.3	2.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.6	7.4	9.9	13.0	15.3	18.1	19.6	19.7	18.0	14.3	10.4	7.2		
		MCDB	9.5	13.1	19.1	23.2	25.1	30.5	33.1	32.0	29.8	23.6	15.6	12.6		
	2%	WB	4.1	5.7	8.5	11.5	14.1	17.0	18.6	18.4	16.3	12.7	9.1	5.8		
		MCDB	7.8	10.3	16.7	19.3	24.0	29.1	31.7	30.4	28.1	22.5	13.6	10.3		
	5%	WB	3.0	4.4	7.4	10.5	13.2	16.1	17.9	17.5	15.3	11.8	8.0	4.5		
		MCDB	6.7	8.3	14.5	17.6	22.7	28.2	30.3	29.3	26.9	20.7	12.4	8.4		
	10%	WB	1.9	3.3	6.3	9.6	12.5	15.2	17.1	16.7	14.5	11.0	7.0	3.4		
		MCDB	5.0	6.9	12.5	16.4	21.6	26.9	29.3	28.4	25.5	19.2	11.6	6.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.7	10.0	11.5	11.9	13.5	15.5	15.2	16.3	16.7	14.1	11.0	9.9		
		MCDBR	11.1	12.7	14.9	14.8	16.0	17.7	17.5	18.0	18.6	16.9	14.2	12.7		
	5% WB	MCWBR	6.8	7.6	7.6	6.6	6.2	6.3	5.9	6.6	7.8	7.9	7.5	7.6		
		MCWBR	9.8	10.6	12.8	12.6	14.6	16.2	15.9	16.2	17.3	15.3	11.1	11.1		
(40) Clear-Sky Solar Irradiance	taub		0.303	0.341	0.395	0.472	0.466	0.453	0.491	0.451	0.396	0.391	0.330	0.300		
	taud		2.326	2.217	2.072	1.901	1.956	2.013	1.928	2.034	2.166	2.136	2.300	2.361		
	Ebn at Noon		883	885	862	814	825	834	798	823	851	814	840	862		
	Edh at Noon		95	120	153	192	185	174	188	166	137	129	97	86		
(44) All-Sky Solar Radiation	RadAvg		2.65	3.58	4.54	5.36	6.66	7.93	7.75	7.17	5.91	4.13	2.75	2.24		
	RadStd		0.21	0.40	0.34	0.45	0.41	0.42	0.24	0.20	0.24	0.28	0.33	0.28		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.66	N/A	N/A	+1.18	+0.83	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)		+0.40	N/A	N/A	+0.99	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page